

Thomas Blaauw Barbosa

Lead Software Engineer | Backend & Systems Architecture

818-306-6834 · thblaaauw@gmail.com · GitHub: thblaaauw.github.io

SUMMARY

Lead software engineer specializing in Python, AWS, distributed systems, and backend architecture. Design and operate production systems serving 80K+ users and 4M+ daily requests, with demonstrated impact in automation, reliability, data synchronization, and operational efficiency. Experienced owning systems from architecture and implementation through CI/CD, deployment, and production operations.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL

Backend: FastAPI, REST APIs, Microservices, Distributed Systems, Async/Concurrent Programming, Multiprocessing, Multithreading

AWS: Lambda, ECS Fargate, EC2, DynamoDB, Aurora PostgreSQL, RDS, SQS, EventBridge, S3, Cognito

Engineering: Docker, CI/CD, Alembic, Unit Testing, Database Migrations, System Architecture

Integrations: Salesforce, Boto3, Simple Salesforce, Event-Driven Architecture, ETL

EXPERIENCE

Inter-Con Security Systems— Pasadena, CA

Lead Software Engineer | 2025–Present

Python Developer | 2022–2025

Junior Python Developer | 2021–2022

Architected and launched an automated shift-scheduling platform on ECS Fargate, automatically filling approximately 80% of company-wide unscheduled shifts, reducing operational costs and officer turnover.

Own the company's primary FastAPI REST platform, serving 80,000+ users and processing 4M+ requests/day; architect and maintain the backend foundation supporting additional company services.

Built an event-driven AWS↔Salesforce ETL platform using SQS, Lambda, EventBridge, Boto3, and Simple Salesforce, reducing synchronization latency from ~2 seconds to <0.4 seconds through asynchronous and concurrent processing.

Architected a proactive missed-shift notification service using push notifications and email, enabling managers to intervene before critical deadlines and reducing incomplete work orders by 60%.

Led a company-wide logging and observability initiative, improving production visibility and reducing downtime recovery time by >90% while cutting typical bug-fix turnaround from 2–3 weeks to 2–3 days.

Designed and built an internal new-hire onboarding platform using Python, EC2, PostgreSQL/Aurora, and Alembic, with database migration testing, unit tests, and CI/CD integrated into the deployment pipeline.

Designed and maintained the organization's core DynamoDB data-access layer, used across production services including the primary REST API; provides relational-style abstractions while retaining DynamoDB's NoSQL scalability and flexibility.

Design and evolve backend architecture across serverless, containerized, monolithic, and microservice systems, with ownership spanning architecture, implementation, testing, CI/CD, deployment, and production operations.

Garage Game Studio — Remote

Software Engineer | 2020–2021

Shipped The Adventures of Chicken Biscuit, an Android mobile game; implemented gameplay systems including collision detection, enemy AI, projectile mechanics, player movement, and UI.

Led 3 programmers, coordinating development priorities and weekly implementation.

EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science in Computer Science · December 2020

Dean's List of Excellence · 2018–2020